Prompt Engine Web — Project Context

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Project summary

Prompt Engine Web is the web version of our Prompt Engine mobile app. It adds two things the mobile app does not have yet: a **prompt marketplace** where creators sell high-quality image and video prompts, and **integrations** with AI generation tools and social media platforms.

- **Why the marketplace:** many creators on Instagram promote and sell image and video prompts through reels and ads. They currently sell through DMs, link-in-bio pages and file downloads. We give them a proper storefront, protected delivery and payouts, and give buyers previews, reviews and one-click generation.
- **Why integrations:** users should run a prompt in their AI tool of choice and publish the result to social media without leaving the app.
- **How we build it:** in phases, backend first. One shared backend serves the web app, the existing mobile app and outside integrations.

This document is the working context for the project. Anyone joining the team, or any AI assistant helping on it, should read it first.

Starting point

The web app keeps feature parity with the mobile app's prompt engine and then goes further. What we carry over from the original Prompt Engine plan:

- Prompt library with folders, tags, search and templates with variables
- Prompt editor and multi-model playground
- Version history, diffs and rollback
- Test sets for scoring prompt changes
- Team workspaces with roles
- Prompt API for calling published prompts from other apps

Rule: web and mobile share one backend and one account system. A prompt saved, bought or published on one platform appears on the other immediately.

To confirm: the mobile app's current stack, backend and database. If it already has a backend, Phase 1 extends that backend rather than starting a new one (see Open questions).

Users and roles

One account can hold several roles: a creator who sells prompts is also a buyer.

Role	Who they are	Main things they do
Buyer	Content creators, marketers, designers, hobbyists	Browse, preview, buy prompts; generate with AI tools; post to social media
Creator (seller)	Prompt sellers, often already promoting on Instagram	Build a storefront, list prompts with sample outputs, set prices, track sales, get paid
Prompt engineer	Developers and product teams	Use the library, playground, versions, test sets and Prompt API
Team member	Staff inside a workspace	Work on shared prompts with owner, editor or viewer rights
Admin / moderator	Our staff	Approve creators and listings, handle reports and refunds, manage payouts and fees

Feature modules

The product has five modules. Each one is a separate backend module with its own API group, so they can be built and shipped in phases.

1. Prompt engine core (enhanced)

- Everything listed under Starting point, now on web and mobile
- **Media prompts:** structured fields for image and video prompts, such as subject, style, camera, lighting, aspect ratio, duration, negative prompt and seed
- **Tool-specific variants:** one prompt can store a tuned version per AI tool, because the same prompt behaves differently in each

- **Output gallery:** generated images and videos are saved against the prompt version that made them

2. Prompt marketplace (new)

- **Creator storefronts:** profile, bio, social links, follower-facing shop page with its own short link for Instagram bios
- **Listings:** title, category (image, video, text), target AI tools, sample outputs, price, licence type and tags
- **Protected delivery:** the prompt text stays hidden until purchase; buyers then get it in their library, plus one-click "Generate" through a connected AI tool
- **Selling options:** single prompts, bundles (packs), and creator subscriptions
- **Payments:** checkout, platform commission, creator balance, payout requests and refunds
- **Trust:** ratings and reviews from verified buyers, report button, moderation queue, duplicate-prompt detection
- **Discovery:** search, filters by tool and style, trending, new, and featured collections
- **Creator dashboard:** views, conversion, sales, earnings and top listings
- **Promo kit:** auto-generated reel covers, captions and short links to help creators promote listings on Instagram and TikTok

3. AI tool integrations (new)

- **Provider adapter layer:** one internal interface for text, image and video generation, with one adapter per provider
- **Two connection modes:** platform credits (we pay the provider and charge users) or bring-your-own API key
- **Async jobs:** video and image generation runs as queued background jobs with status updates and webhooks
- **Copy-out mode:** for tools with no public API, the app formats the prompt for that tool and copies it, so users paste it there
- **Target tools:** chosen per phase from the image, video and text tools that offer an official API (see Open questions)

4. Social media integrations (new)

- **Connect accounts** through each platform's official sign-in (OAuth)

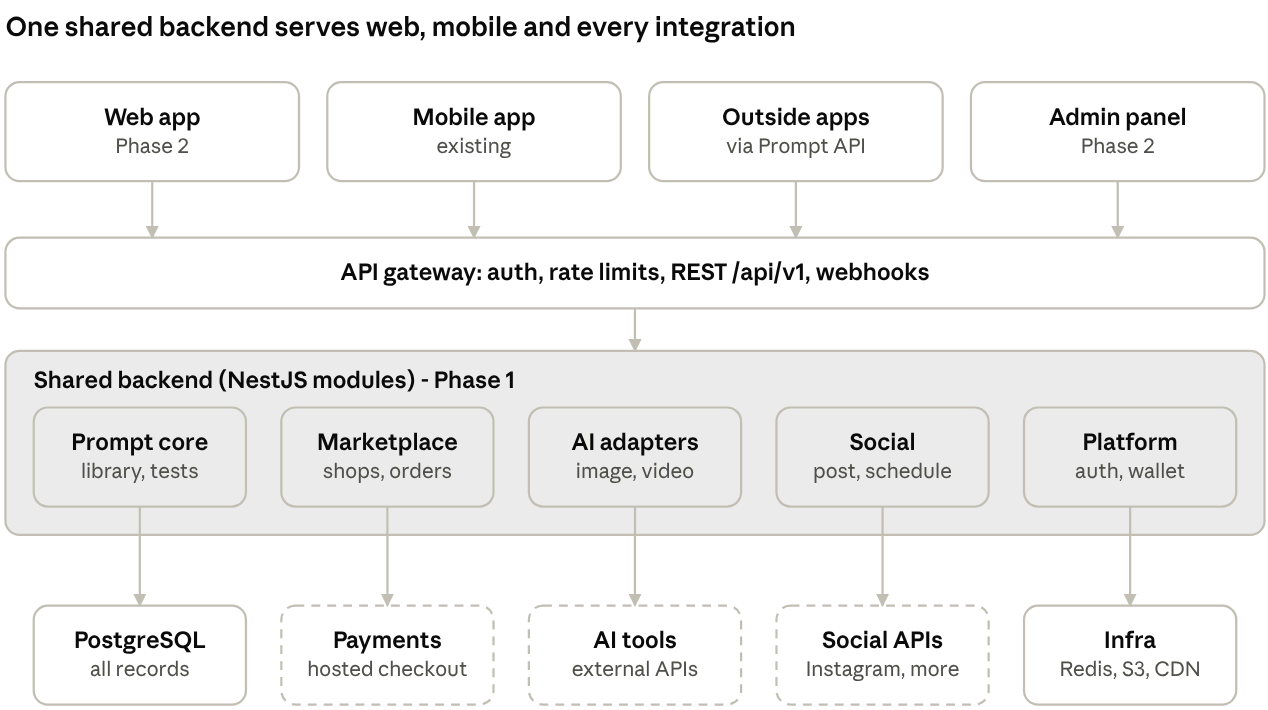
- **Publish or schedule** generated images and videos with caption and hashtags, to platforms whose APIs allow it
- **Share listings:** creators push a listing to their social accounts as a post or story with a buy link
- **Social sign-in:** sign up with Google, Apple, Facebook or similar
- **Basic analytics:** clicks and sales that came from each social link
- **Target platforms:** Instagram and Facebook first, then TikTok, YouTube, Pinterest and X, each subject to its API rules

5. Platform services (shared)

- Authentication, profiles and roles
- Wallet and credits for AI generation
- Media storage and CDN for samples and outputs
- Notifications: in-app, email and mobile push
- Admin panel: users, creators, listings, orders, payouts, reports, fees and platform settings
- Audit log and usage tracking

Architecture

One backend, built first, serves every client. The web app, mobile app, admin panel and outside developers all use the same API.



system architecture · clients, backend modules, services

AI tools, social platforms and payments sit behind adapters. Adding or swapping one does not touch the rest of the system.

Data model

These are the main entities for Phase 1. Field lists are a starting point, not a final schema.

Entity	Key fields	Module
User	id, email, name, avatar, roles, auth providers, locale, status	Platform
Workspace	id, name, owner, members with roles, plan	Platform
Prompt	id, owner or workspace, title, type (text, image, video), visibility, tags	Core
PromptVersion	id, prompt, body, variables, media fields, tool variants, model settings, created by	Core

Entity	Key fields	Module
TestSet / TestRun	inputs, expected outputs, scores, version tested	Core
Generation	id, prompt version, tool, parameters, status, output media, cost, user	AI integrations
ToolConnection	user, provider, mode (credits or own key), encrypted key, status	AI integrations
CreatorProfile	user, shop handle, bio, social links, verification, payout details	Marketplace
Listing	id, creator, linked prompt version, title, category, target tools, price, licence, status	Marketplace
ListingMedia	listing, sample image or video, tool used, order	Marketplace
Bundle / Subscription	creator, included listings, price, billing period	Marketplace
Order / OrderItem	buyer, items, amounts, commission, payment status, gateway reference	Marketplace
Entitlement	buyer, listing or bundle, granted at, source order	Marketplace
Review / Report	author, listing, rating or reason, text, moderation status	Marketplace
Payout / LedgerEntry	creator, amount, currency, type, status	Marketplace
SocialAccount	user, platform, account id, tokens (encrypted), scopes, expiry	Social
SocialPost	user, account, media, caption, scheduled time, status, platform post id	Social
Wallet / CreditTransaction	user, balance, top-ups, spend per generation	Platform
Notification / AuditLog	user, type, payload, read state / actor, action, target, time	Platform

An **Entitlement** is what unlocks a purchased prompt. Prompt text is only returned by the API when the requester owns the prompt or holds an entitlement for it.

API surface

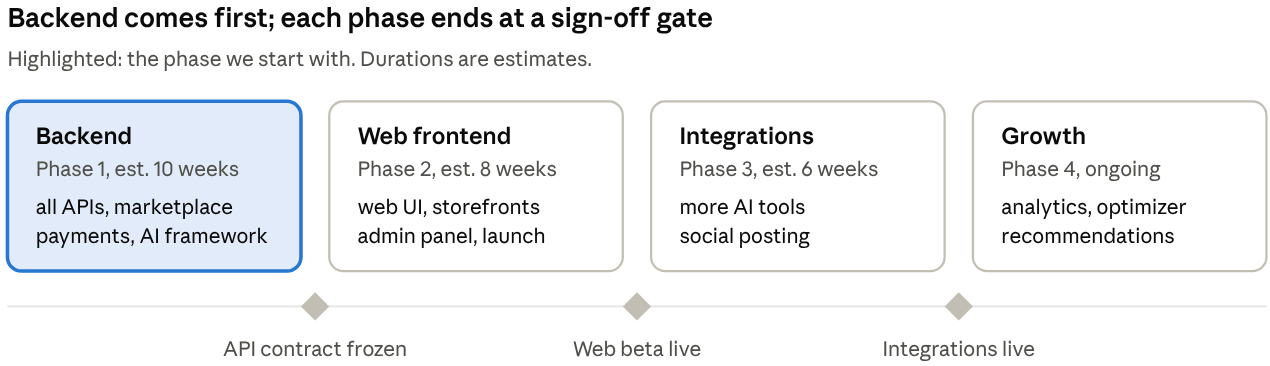
The backend exposes a versioned REST API (/api/v1) used by the web app, the mobile app and outside developers. Long-running work reports back through webhooks and real-time events.

Group	Example endpoints	Phase
Auth	POST /auth/register, /auth/login, /auth/refresh, /auth/oauth/{provider}	1
Users and workspaces	GET /me, PATCH /me, /workspaces, /workspaces/{id}/members	1
Prompts	/prompts, /prompts/{id}/versions, /prompts/{id}/diff, /prompts/{id}/rollback	1
Tests	/test-sets, POST /test-sets/{id}/run	1
Prompt API (public)	POST /run/{promptId} with variables, API keys per workspace	1
Marketplace browse	GET /listings, /listings/{id}, /creators/{handle}, /search	1
Creator tools	/creator/profile, /creator/listings, /creator/bundles, /creator/stats	1
Orders and payments	POST /checkout, /orders, /entitlements, payment gateway webhooks	1
Payouts	/creator/balance, POST /creator/payouts	1
Reviews and reports	/listings/{id}/reviews, POST /reports	1
Generation	POST /generations, GET /generations/{id}, /tool-connections	1 (framework), 3 (more tools)
Social	/social/accounts, POST /social/posts, /social/posts/{id}/schedule	3
Wallet	/wallet, POST /wallet/top-up	1
Admin	/admin/users, /admin/listings/{id}/approve, /admin/payouts, /admin/reports	1

The API is documented with OpenAPI (Swagger) from day one, so the web and mobile teams build against the same contract.

Phased delivery

We build the backend first so the web app and the mobile app can both build against a finished API.



delivery roadmap · 4 phases, 3 gates

Phase 1: Backend

- Auth, users, workspaces, roles and wallet
- Prompt core APIs, including media prompt fields and tool variants
- Marketplace APIs: creators, listings, bundles, search, checkout, entitlements, reviews, reports, payouts and ledger
- Payment gateway integration and webhooks
- AI adapter framework with one or two image tools and one video tool
- Admin APIs, OpenAPI docs, tests, staging environment
- Mobile app connected to the new marketplace endpoints

Phase 2: Web frontend

- Web app: library, editor, playground, marketplace browse and checkout, creator storefront and dashboard
- Admin panel for moderation, orders and payouts
- Public launch of the web app

Phase 3: Integrations

- More AI tool adapters, chosen by user demand
- Social account connection, publish and schedule, listing share, link tracking
- Creator promo kit for reels and posts

Phase 4: Growth

- Analytics for creators and admins
- AI prompt optimizer and personalized recommendations
- Creator subscriptions, affiliate links and featured placements

New requests are added as a new module or a later phase, then this document is updated.

Rules and constraints

Security

- All third-party API keys and social tokens are encrypted at rest and never returned to the client after saving.
- Role-based access checks on every endpoint; workspace data never leaks across workspaces.
- Rate limits on login, generation, checkout and the public Prompt API.

Prompt protection

- Purchased prompt text is only served to owners and entitled buyers, never in listing pages, search results or page source.
- Watermark buyer copies with a hidden buyer ID to trace leaks; add a report flow for stolen or copied prompts.
- Sample outputs are stored separately from prompt text and can be watermarked.

Payments

- Never store card details; use a payment gateway with hosted checkout.
- Keep a double-entry ledger for sales, commission, refunds and payouts, so every creator balance can be traced.
- The gateway must work for a Pakistan-based business and support international buyers. Stripe does not currently onboard Pakistan-registered businesses, so local and international options need checking (see Open questions).

Platform and content policies

- Use only official, documented APIs for AI tools and social platforms. No scraping or unofficial automation, which risks account bans.
- Each social platform restricts what can be posted through its API (for example, account type and media format). Each integration is checked against current rules when it is built.

- Check each AI tool's terms on commercial use of outputs, and state licence terms clearly on every listing.
- Moderate listings for adult content, real-person deepfakes, copyrighted characters and brand misuse before they go live.

Performance

- API p95 response under 300 ms for reads; generation jobs report status within 2 seconds of a change.
- Media served through a CDN; images resized on upload.

Working conventions

These defaults apply unless the mobile app's existing stack makes a different choice better.

Area	Default
Backend	Node.js with NestJS, TypeScript, REST + OpenAPI, modular monolith (one module per feature area)
Database	PostgreSQL with Prisma ORM; migrations in the repo
Queue and cache	Redis with BullMQ for generation, posting, payouts and emails
Storage	S3-compatible object storage with CDN for media
Web frontend (Phase 2)	Next.js (React), TypeScript, Tailwind CSS
Auth	JWT access + refresh tokens, OAuth for social sign-in
Testing	Jest unit tests per service, end-to-end API tests for every endpoint group
DevOps	Docker, GitHub Actions CI/CD, staging and production environments

Repo layout (monorepo)

```
apps/
  api/          NestJS backend
  web/          Next.js web app (Phase 2)
  admin/        Admin panel (Phase 2)
packages/
  shared-types/ API types shared with web and mobile
  ai-adapters/  One adapter per AI provider
  social-adapters/ One adapter per social platform
docs/           This context, API notes, decisions
```

Rules for the team and AI coding assistants

- Read this document before starting any task; update it when a decision changes.
- Each feature module owns its tables and endpoints; modules talk through service interfaces, not each other's tables.
- New AI tools or social platforms are added as adapters only, never as special cases in core code.
- Every endpoint has input validation, an auth guard, an OpenAPI description and a test.
- Money is stored as integers in the smallest currency unit, with the currency code beside it.
- Secrets come from environment variables, never from code or commits.

Open questions

These need answers before Phase 1 starts.

- [ ] What is the mobile app's current stack and backend? Do we extend it or build the shared backend fresh and migrate?
- [ ] Which payment gateway(s) for a Pakistan-based business selling to local and international buyers (for example local wallets and cards, plus an international processor)?
- [ ] Platform commission rate and creator payout schedule and minimum.
- [ ] Pricing currency: PKR, USD, or both?
- [ ] Which AI tools in the first release? Confirm each has an official API and allows commercial use.
- [ ] Which social platforms after Instagram and Facebook, and in what order?
- [ ] Generation billing: platform credits, bring-your-own-key, or both at launch?
- [ ] Who can sell: open sign-up, or approved creators only at launch?

- [ ] Licence types to offer on listings (personal use, commercial use, extended).